CS410 Midterm 3 Part (a)

30 December 2022

I. Multiple Choice Questions (50 points)

1. Which of the following is not correct for Turing machines?

a) A Turing machine can both write on tape and read from it.
b) The head can move both to the left and to the right.
c) The tape is finite.
d) The states for rejecting and accepting take effect immediately.
e) All of them correct.

Given the following Turing machine description, answer questions 2-4 accordingly. M_1 for the language $B = \{w\#w | w \in 0, 1^*\}$ as:
On input string w;

a. Zig-zag across the tape to corresponding position on either side of the # symbol to check whether these positions contain the same symbol. If they do not, or if no # is found, reject. Cross off symbols as they are checked to keep track of which symbols correspond.
b. When all symbols to the left of the # have been crossed off, check for any remaining symbols to the right of the #. If any symbols remain, reject; otherwise accept.

2. Which of the following string is possible while iterating the Turing machine on input 010011#010011?

a) 0x0011#0x0011
b) 010011x010011
c) xxx011#xxx011
d) xxx01x#xxx01x
e) All of them possible

3. Which of the following string is on the tape just before the Turing machine accepts on input 010011#010011?

a) xxxxxxxxxxxxx
b) 010011x010011
c) xxx011#xxx011
d) xxxxx1#xxxxx1
e) xxxxxx#xxxxxx

4. Which of the following string is on the tape just before the Turing machine rejects on input 010011#01001111?

a) xxxxxxxxxxxxx11
b) 010011x01001111
c) xxx011#xxx01111
d) xxxxx1#xxxxx111
e) xxxxxx#xxxxxx11

5. Which of the following string is a possible initial configuration for a Turing machine?

a) 101101101
b) $q_0$1011$q_2$01101
c) $q_0$101101101
d) All of the above
e) None of the above

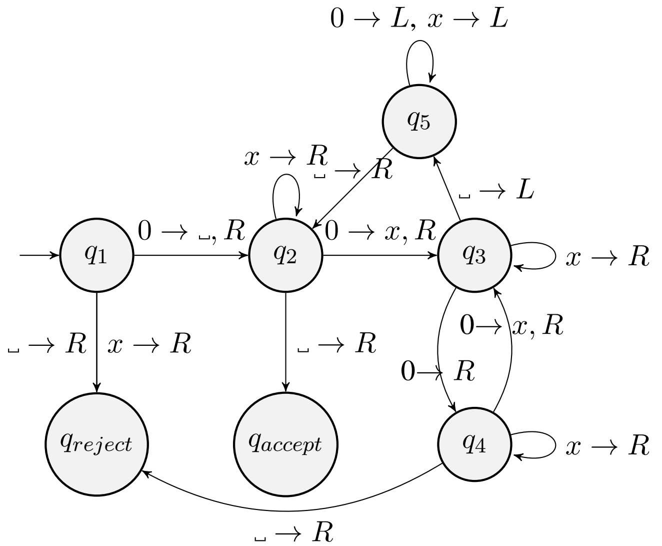
6. If we apply the state transition function $\delta(q_i, b) = (q_j, c, L)$ to the configuration uaq_ibv, which configuration will we have next?

a) uq_jacv
b) uaq_jcv
c) uaq_jbcv
d) $uacq_jv$
e) None of the above

7. If we apply the state transition function $\delta(q_i, b) = (q_j, c, R)$ to the configuration uaq_ibv, which configuration will we have next?

a) uq_jacv
b) uaq_jcv
c) uaq_jbcv
d) $uacq_jv$
e) None of the above

Given the following Turing machine TM_1. Answer questions 8-11 accordingly.



8. What is Q?
  a) $\{q_1, q_2, q_3, q_4, q_5\}$
  b) q_1
  c) q_{accept}
  d) $\{q_1, q_2, q_3, q_4, q_5, q_{accept}, q_{reject}\}$
  e) $\{q_{accept}, q_{reject}\}$

9. What is Σ?
  a) $\{0\}$
  b) $\{0, x\}$
  c) $\{0, x, \sqcup\}$
  d) $\{0, x, \sqcup, L\}$
  e) $\{0, x, \sqcup, L, R\}$

10. What is τ?
  a) $\{0\}$
  b) $\{0, x\}$
  c) $\{0, x, \sqcup\}$
  d) $\{0, x, \sqcup, L\}$
  e) $\{0, x, \sqcup, L, R\}$

11. What is the language generated by TM_1?
  a) $L = \{0^n \mid n \geq 0\}$
  b) $L = \{0^{2^n} \mid n \geq 0\}$
  c) $L = \{0^{n^2} \mid n \neq 0\}$
  d) $L = \{0^n 1^n \mid n \geq 0\}$
  e) None

12. How do we implement stay put S with L and R?
  a) S can not be implemented
  b) First Left move, then Right move
  c) First Right move, then Left move
  d) b and c is correct
  e) None

13. In how many possible ways can the heads of a k tape multi-tape Turing machine move?
  a) k
  b) k!
  c) 2^k
  d) k^3
  e) k^2

14. Let say we have 2-tape Turing machine with tape contents [1 0 0 1 ␣] and [a b a ␣]. Let also say that currently the first tape head shows the second 1, and the second tape head shows b. What will the single tape version of this 2-tape Turing machine?
  a) [1 0 0 1 a b a]
  b) [1 0 0 1 ␣ a b a ␣]
  c) [1 0 0 1 a b a ␣]
  d) [1 0 0 $\dot{1}$ a $\dot{b}$ a ␣]
  e) [1 0 0 $\dot{1}$ # a $\dot{b}$ a # ␣]

15. Which tapes do we utilize in simulation the Nondeterministic TM with multi-tape deterministic TM?
  a) input tape, address tape
  b) input tape, simulation tape
  c) address tape, simulation tape
  d) input tape, address tape, simulation tape
  e) None of the above

16. Which of the following languages is decidable?
  a) A = {<G, w> | G is a DFA that accepts input string w}
  b) B = {<G, w> | G is an NFA that accepts input string w}
  c) C = {<R, w> | R is a regular expression that generates string w}
  d) D = {<G> | G is a DFA and L(A) = ∅ }
  e) All of the above

17. Which of the following languages is decidable?
  a) A = {<A, B> | A abd B are DFA, L(A) = L(B)}
  b) B = {<G, w> | G is a CFG that generates string w}
  c) C = {<G> | G is a DFA and L(G) = ∅}
  d) Every context free language
  e) All of the above

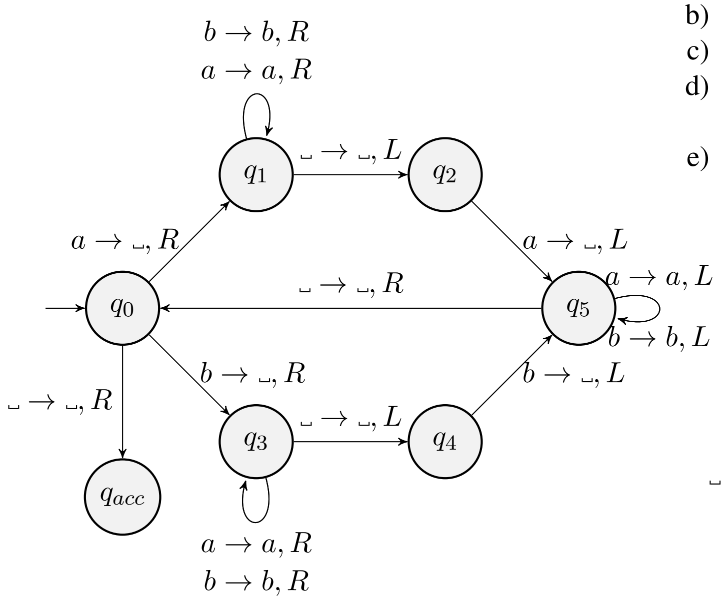
18. Which of the following sets is the largest?
  a) Regular languages
  b) Context-free languages
  c) Decidable languages
  d) Turing-recognizable languages
  e) All of them equal in size

19. Which of the following languages is not Turing-Recognizable?

a) Complement of a Turing-recognizable language
b) Complement of a decidable language
c) Union of two decidable languages
d) Concatenation of two decidable languages
e) All of the above

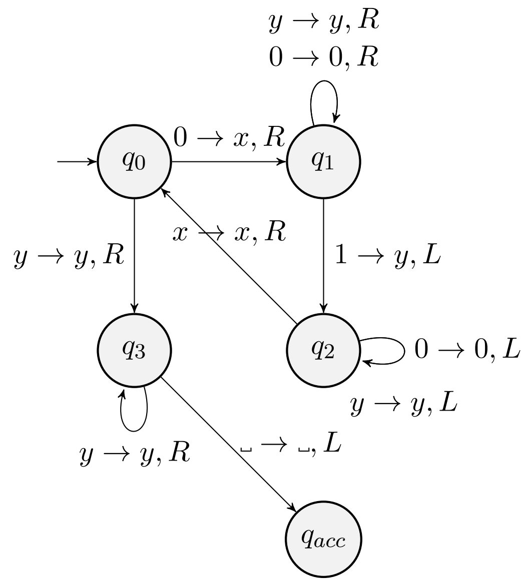
20. Which of the following languages is not countable?

a) N
b) Z
c) Q
d) R
e) All of them are countable



21. Which of the following languages is depicted in the TM given above?

a) L = {w | Number of 1's in w is a multiple of 3 }
b) L = { 0^n1^n | n $\geq$ 1}
c) Language of palindromes of even length
d) Given the input as 1^* doubles the number of symbols.
e) None of the above



22. Which of the following languages is depicted in the TM given above?

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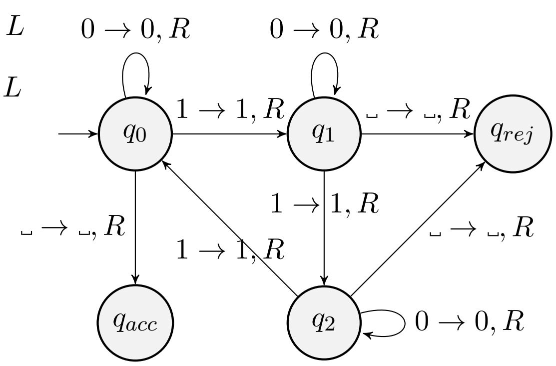

23. Which of the following languages is depicted in the TM given above?

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b) L = { 0^n1^n | n $\geq$ 1}
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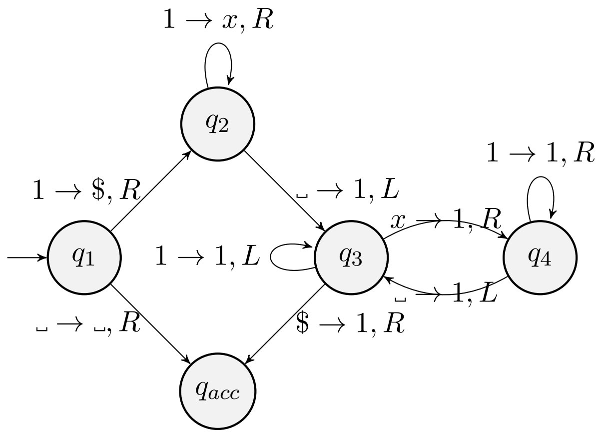

24. Which of the following languages is depicted in the TM given above?

a) L = $\{$w $|$ Number of 1's in w is a multiple of 3 $\}$
b) L = $\{ 0^n1^n$ $|$ n $\geq 1\}$
c) Language of palindromes of even length
d) Given the input as 1^* doubles the number of symbols.
e) None of the above

25. What is the name of the instructor?

a) Ali Cengiz Beğen
b) Hasan Sözer
c) Kübra Kalkan Çakmakçı
d) Olcay Taner Yıldız
e) Reyhan Aydoğan
f) Sedat Özer